AGGREGATE AND GROUPING FUNCTIONS

1. Create table and insert values

Firstly, Creating the table Cloud usage student as we are finding the cost of each services using aggregate function and grouping along with having functions.

SQLQuery5.sql...\Krishy (54))*

```
5   CREATE TABLE cloud_usage(
6       user_id VARCHAR(20),
7       resource_type VARCHAR(20),
8       usage_hours FLOAT,
9       cost FLOAT,
10      usage_date DATE
11  );
12
13  INSERT INTO cloud_usage VALUES
14  ('U001', 'compute', 120.5, 2400, '2025-07-01'),
15  ('U002', 'storage', 300.0, 1500, '2025-07-02'),
16  ('U003', 'network', 50.0, 800, '2025-07-03'),
17  ('U001', 'storage', 200.0, 1000, '2025-07-04'),
18  ('U004', 'compute', 400.0, 5000, '2025-07-05'),
19  ('U002', 'network', 100.0, 1200, '2025-07-06'),
20  ('U005', 'compute', 50.0, 600, '2025-07-07'),
```

100 % No issues found

Results Messages

	user_id	resource_type	usage_hours	cost	usage_date
1	U001	compute	120.5	2400	2025-07-01
2	U002	storage	300	1500	2025-07-02
3	U003	network	50	800	2025-07-03
4	U001	storage	200	1000	2025-07-04
5	U004	compute	400	5000	2025-07-05
6	U002	network	100	1200	2025-07-06
7	U005	compute	50	600	2025-07-07
8	U003	storage	150	750	2025-07-08
9	U004	network	80	900	2025-07-09
10	U001	compute	100	2000	2025-07-10

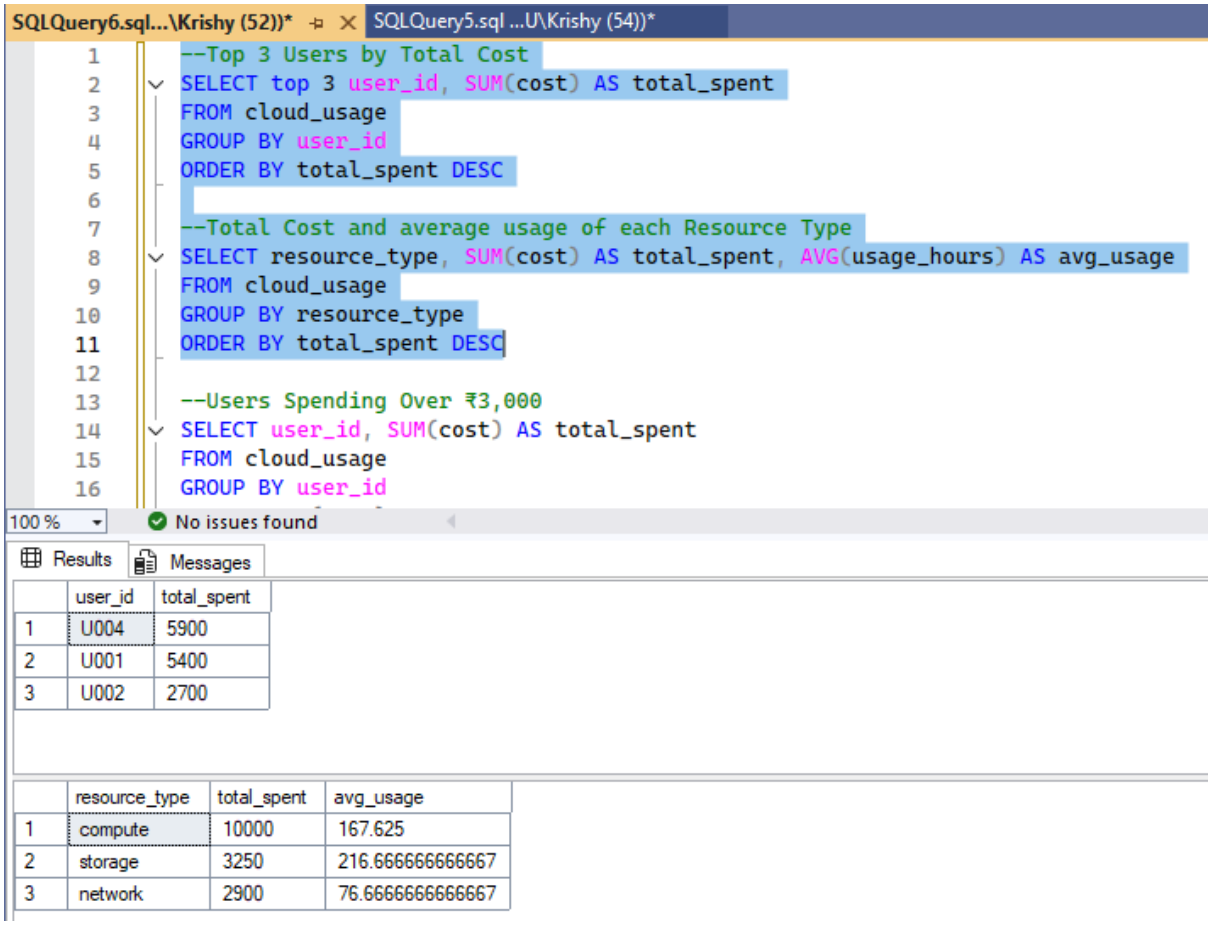
Aggregate functions

Aggregate functions perform a calculation on a set of values (a column) and return a single, summary value. They are typically used with the SELECT statement and are often paired with the GROUP BY and HAVING clause to perform the calculations on subsets of data.

Now we will see one by one…

Sum & Average

- Sum Calculates the total sum of all non-NULL values in a numeric column.
- AVG Calculates the average (mean) value of all non-NULL values in a numeric column.



```sql
--Users Spending Over ₹3,000
SELECT user_id, SUM(cost) AS total_spent
FROM cloud_usage
GROUP BY user_id
HAVING SUM(cost) > 3000;
```

No issues found

Results | Messages

user_id	total_spent
U001	5400
U004	5900

Maximum & Minimum

- Max - Finds the maximum value in a column.
- Min - Finds the minimum value in a column.

```sql
--Users with High Variability in Cost
SELECT
    user_id, MAX(cost) as Max_cost, MIN(cost) as Min_cost,
    MAX(cost) - MIN(cost) AS cost_range
FROM cloud_usage
GROUP BY user_id
HAVING MAX(cost) - MIN(cost) > 2000;

--Users with More Than One Resource Type Used
SELECT
    user_id,
    COUNT(DISTINCT resource_type) AS resource_variety
FROM cloud_usage
GROUP BY user_id
HAVING COUNT(DISTINCT resource_type) > 1;
```

No issues found

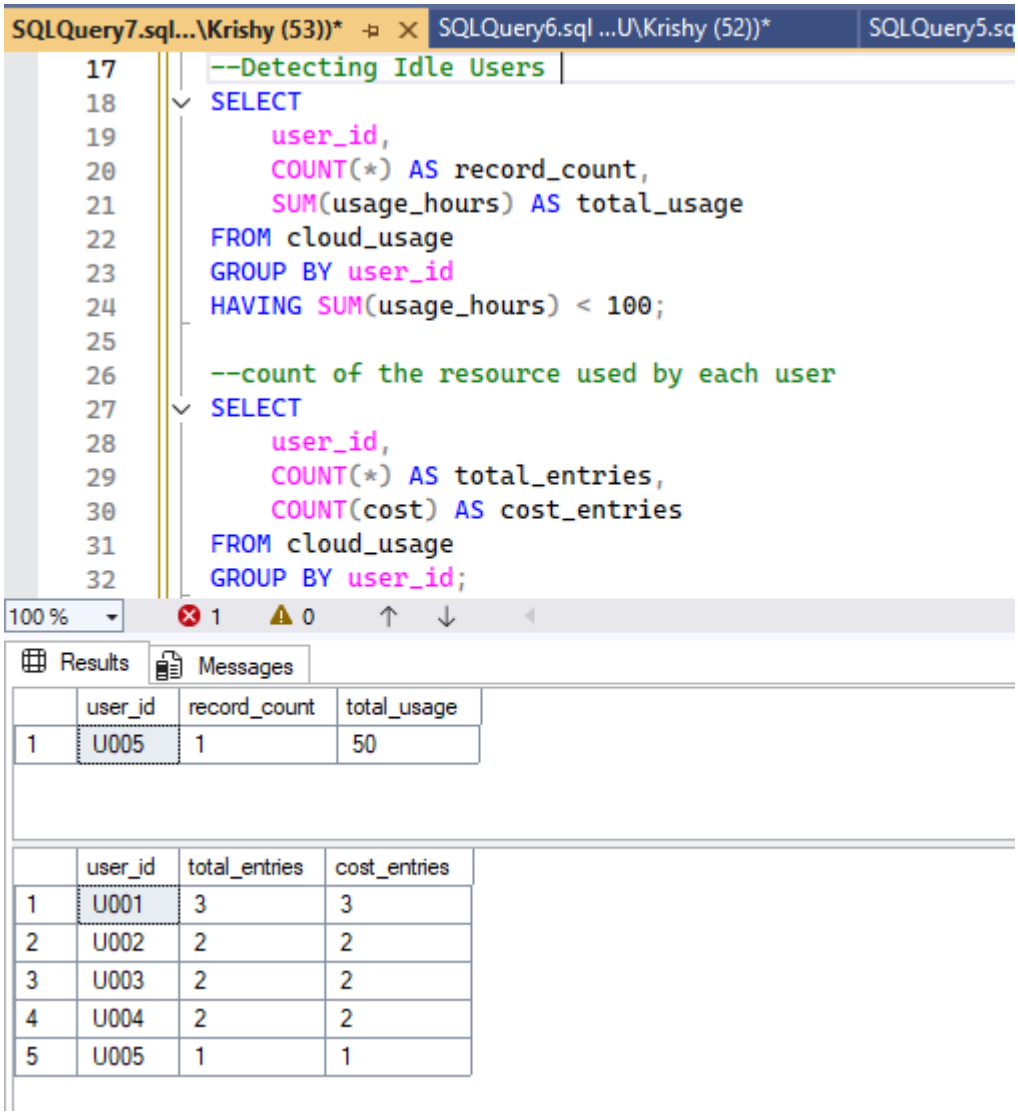
Results | Messages

	user_id	Max_cost	Min_cost	cost_range
1	U004	5000	900	4100

	user_id	resource_variety
1	U001	2
2	U002	2
3	U003	2
4	U004	2

Count()

- COUNT(*): Counts all rows, including those with NULL values. It's the most common way to count all records in a table.
- COUNT(column_name): Counts only the rows where the specified column_name is not NULL.
- COUNT(DISTINCT column_name): Counts the number of unique, non-NULL values in a column.



Thank you!